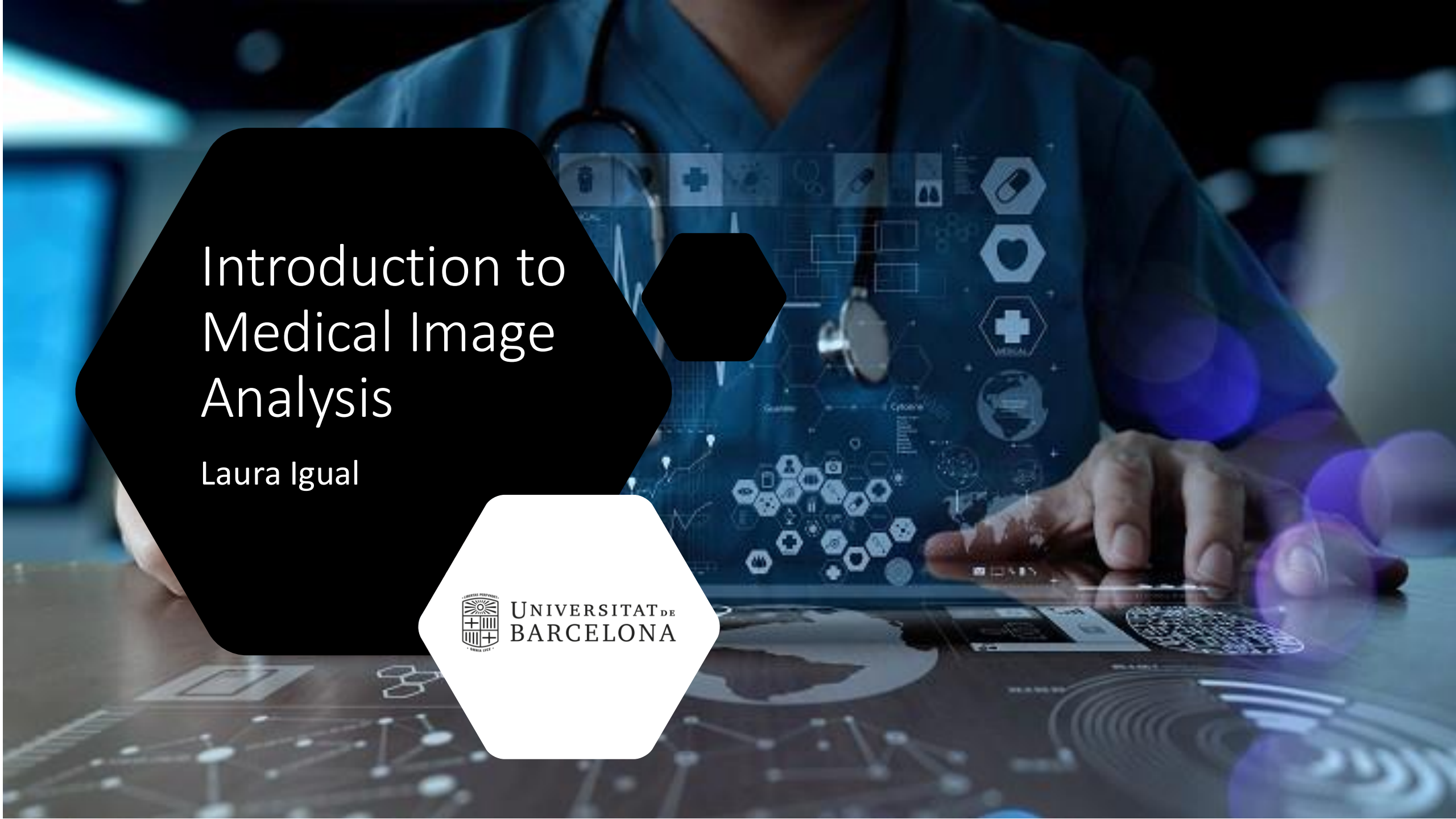


Introduction to Medical Image Analysis

Laura Igual

UNIVERSITAT DE BARCELONA

DSxHealth Syllabus

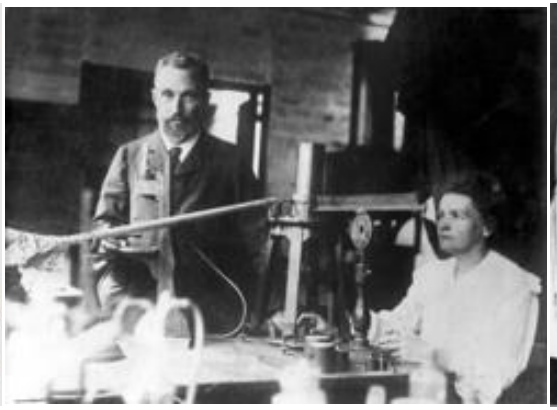
- Introduction to DSxHealth
- Electronic Health Records Exploration
- mHealth Analysis
- **Medical Image Analysis**
- Public Health and Epidemiological Modeling
- DSxHealth Applications

Index of today's class

- Medical Imaging
- Medical Imaging Modalities
- Clinical Problems
- Medical Image Processing and Analysis Tasks
- Datasets and Tools
- References

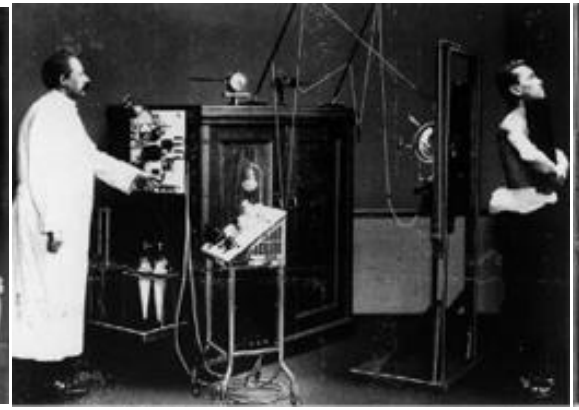
Medical Imaging

- Process of creating visual representations of the interior of a body.
- Supports diagnosis and treatment



First radiography by W. Röntgen in 1896.

Pierre and Marie Curie in the laboratory, 1904.

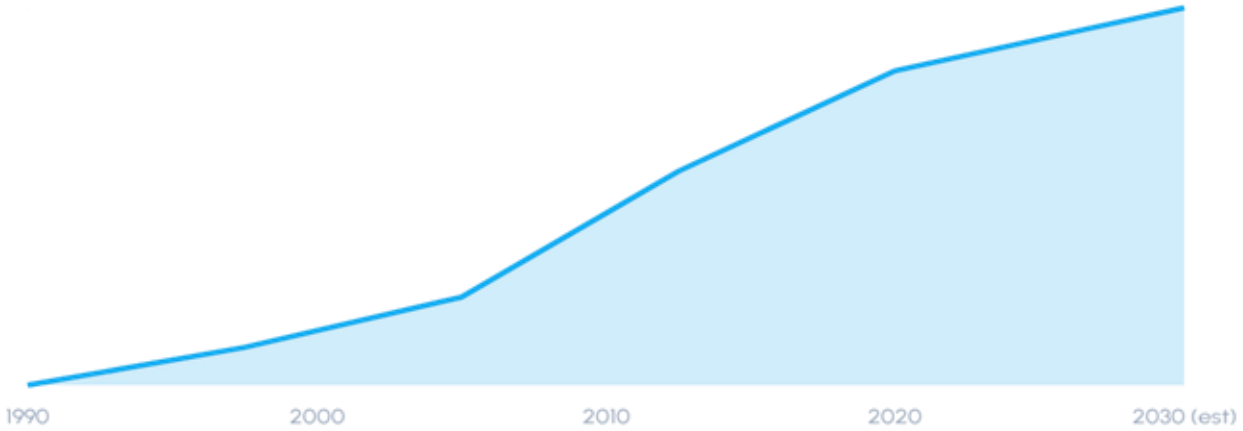


Why Medical Imaging?

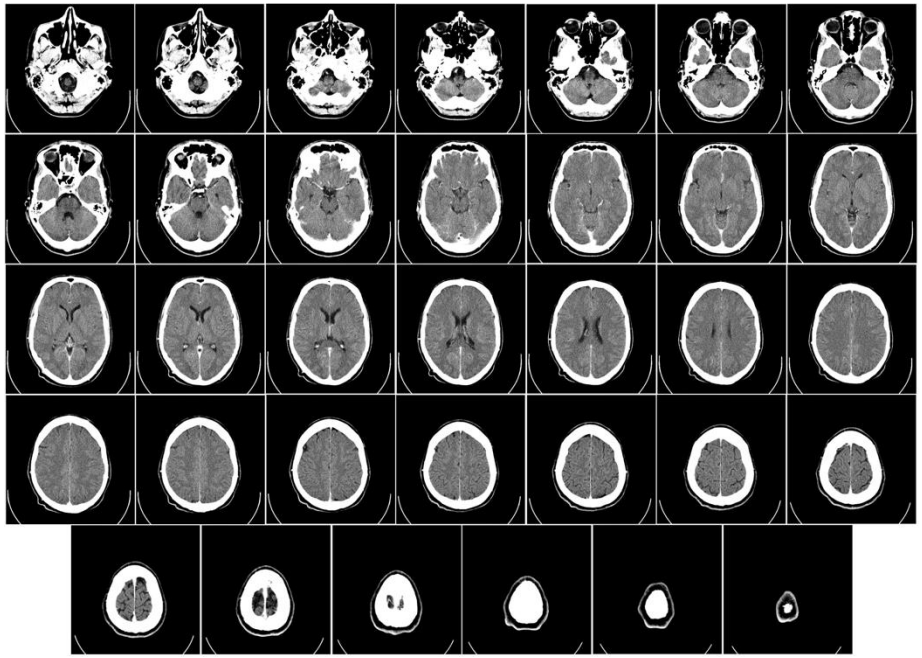
- Key tool in modern medicine
- Non-invasive diagnosis
- Used in almost every specialty

The main challenge

- Huge volume of data
- Manual analysis is slow



Exponential rise in Petabytes of medical image data.



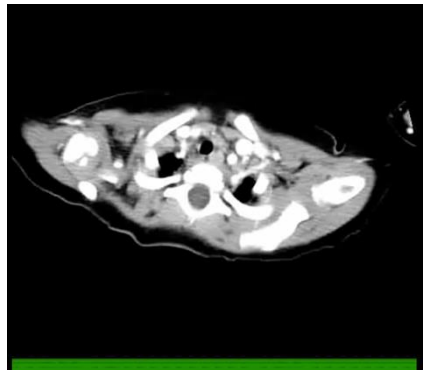
Medical Imaging Modalities

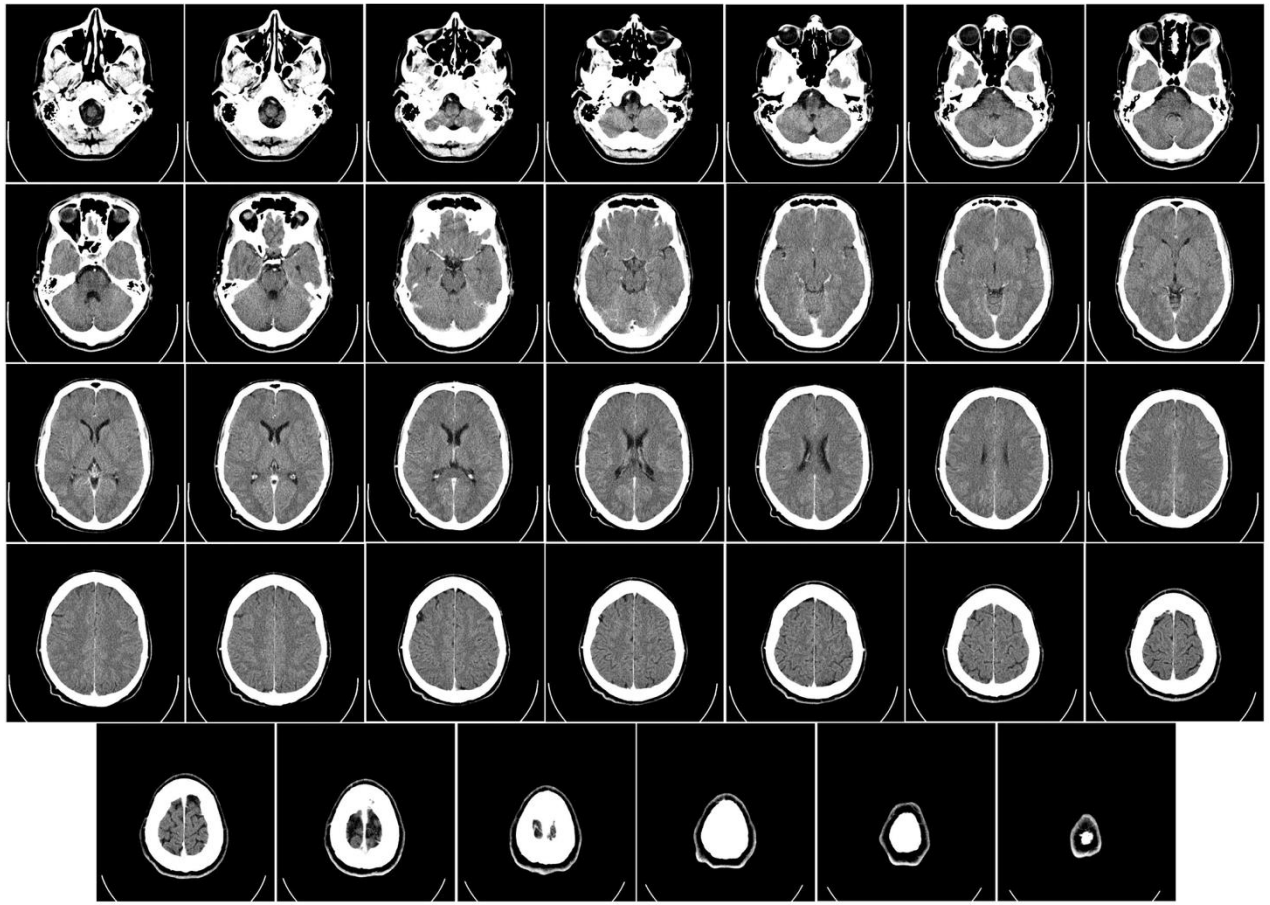
1. Radiography

- Uses X-rays.

2. Computerized Tomography

- Computer-processed combinations of **many X-ray images** taken from different angles to produce **cross-sectional images**.

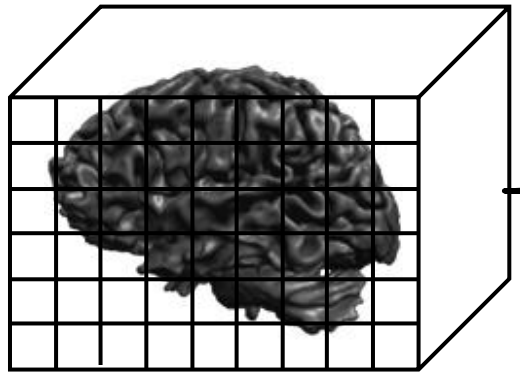


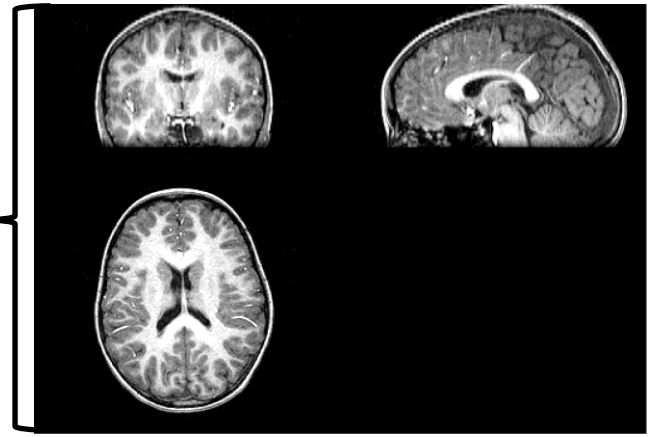


3. Magnetic Resonance Images

- Non-invasive medical imaging technique
- Uses strong magnetic fields and radiofrequency waves
- Excellent soft tissues (brain, muscles, and organs)

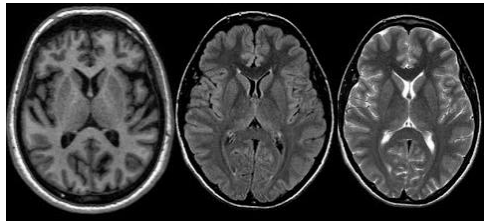
MRI scanner



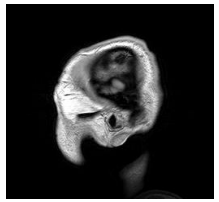


Structura MRI (sMRI) allows to distinguish anatomical structures of different tissues:

- White Matter
- Gray Matter
- Cerebrospinal Liquid.



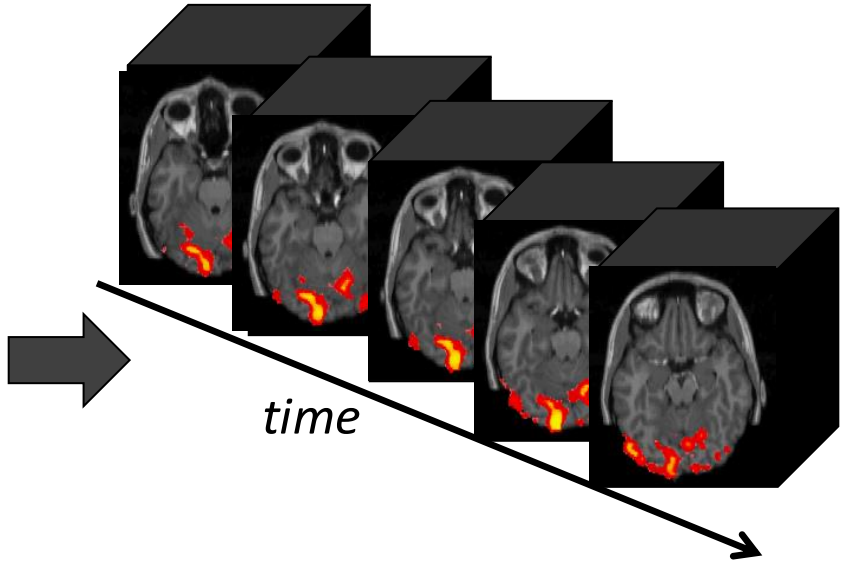
Multiple anatomical MR sequences: T1, T2, FLAIR.

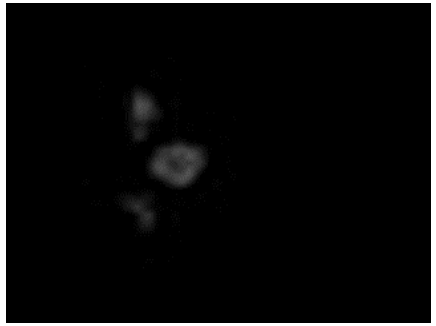


4. Functional Magnetic Resonance Images (fMRI)

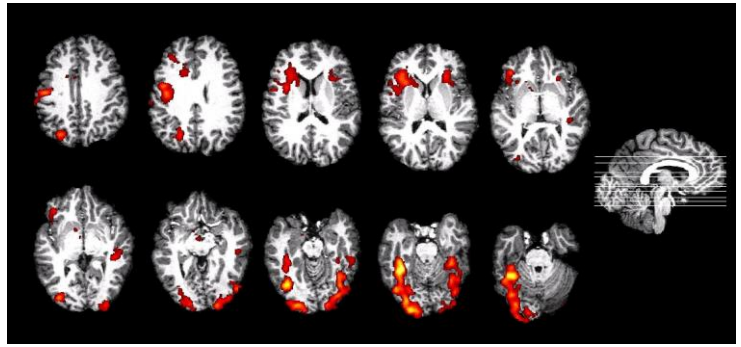
- Measures and maps brain activity by detecting changes in blood oxygenation and flow (the BOLD signal)
- Used to analyse **neural activity** in different regions of the brain over time **in a task or resting.**

MRI scanner





Raw fMRI signal (BOLD)



Processed signal: activated areas in a particular task-based experiment.

Two approaches:

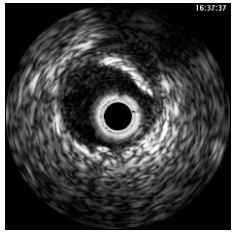
- Task-based fMRI
- Resting state fMRI

5. Ultrasound

- Uses high-frequency sound waves.
- Non-invasive, real-time, and portable.
- Ideal for obstetrics and cardiology.

Carotid ultrasound:

Intravascular ultrasound (IVUS):

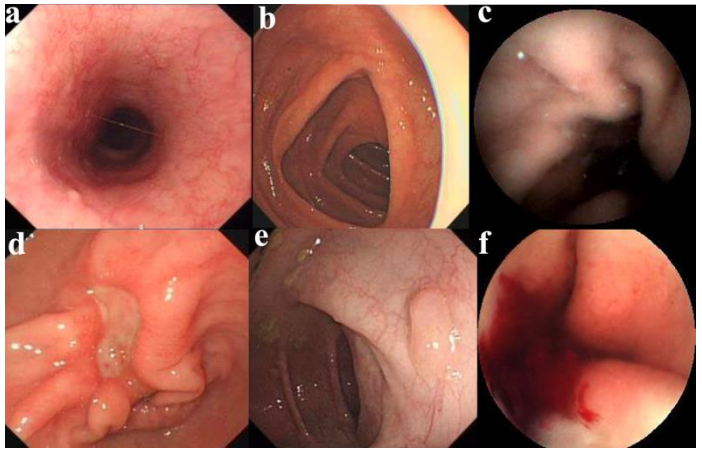


Obstetric ultrasound:

6. Endoscopic image

- Camera inside body.
- Captured using an endoscope
- Direct visualization.

Endoscope.



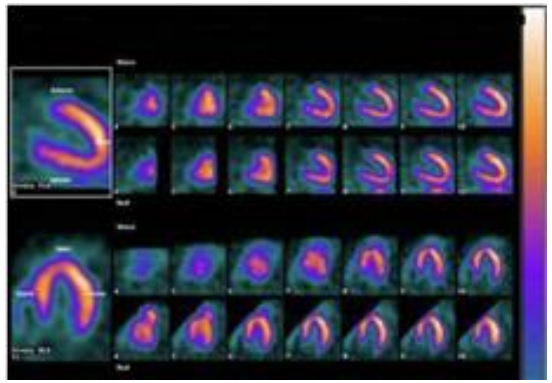
Gastrointestinal endoscopic images: (a) Gastroscopic image of normal esophagus, (b) colonscopic image of normal colorectal, (c) WCE image of normal small intestinal, (d) gastroscopic image of stomach with ulcer, (e) colonscopic image of colorectal with polyp, (f) WCE image of small intestinal with hemorrhage.

Source: W. Du *et al.*, "Review on the Applications of Deep Learning in the Analysis of Gastrointestinal Endoscopy Images," in *IEEE Access*, vol. 7, pp. 142053-142069, 2019, doi: 10.1109/ACCESS.2019.2944676.

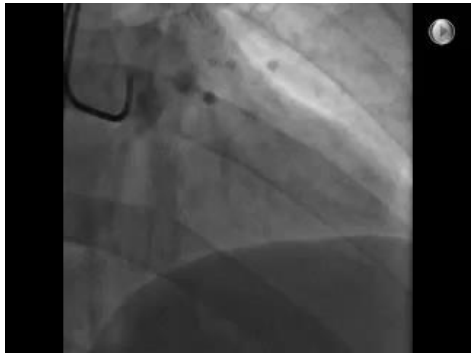
When do we use each modality?

Modality	Best for	Limitation
X-ray	bones	low soft tissue contrast
CT	fast emergency imaging, 3D	radiation
MRI	brain, soft tissue	expensive
Ultrasound	real-time	operator-dependent

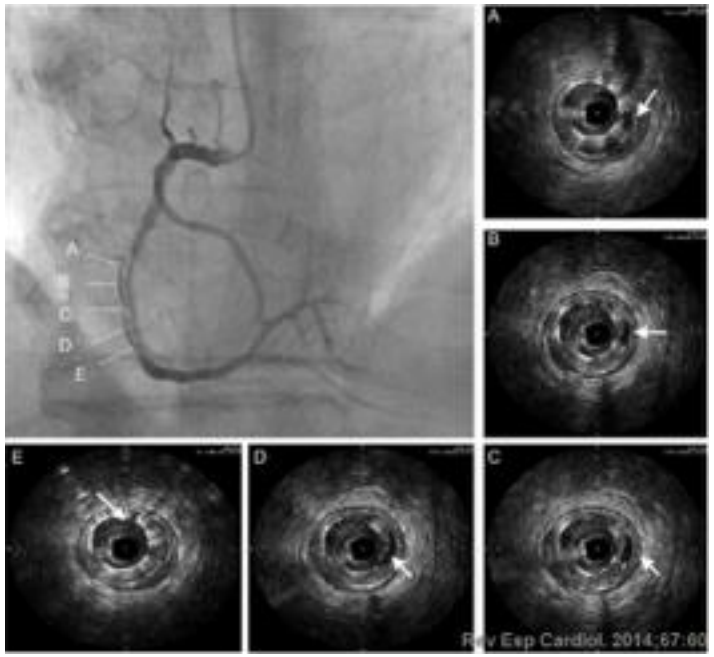
Other Modalities…

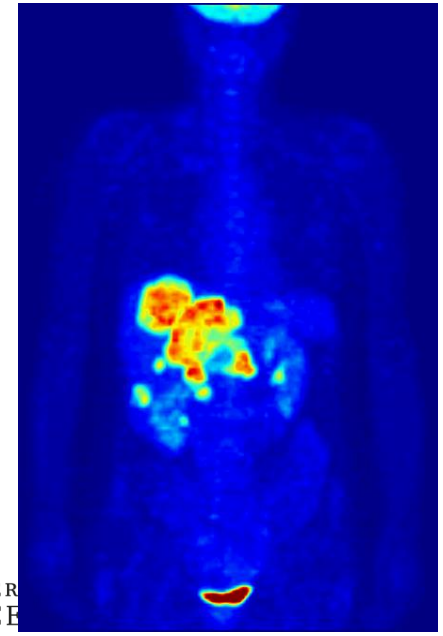


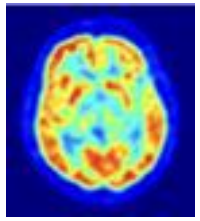
Cardiac Single Photon Emission Computed Tomography (SPECT)



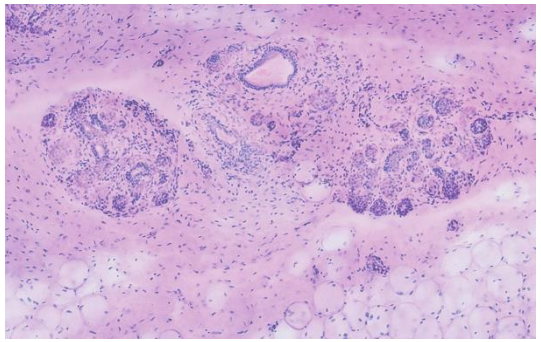
Angiography and intravascular ultrasound



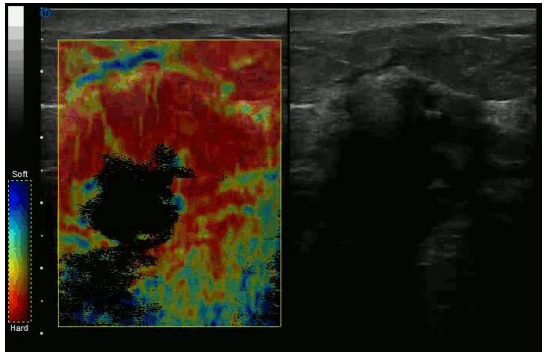




Positron Emission Tomography (PET)



Pathology. Digital slides of tissue sections



Elastography

From images to problems

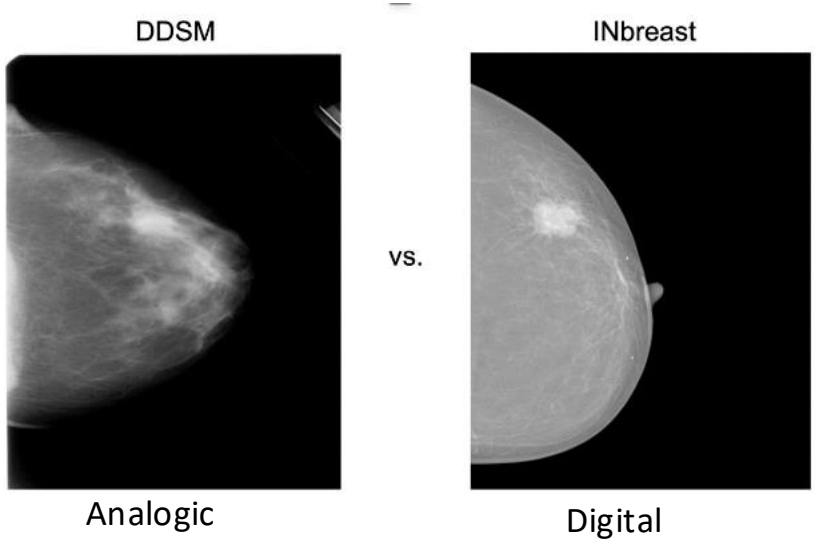
- We have many images
- But, what do we want to solve?

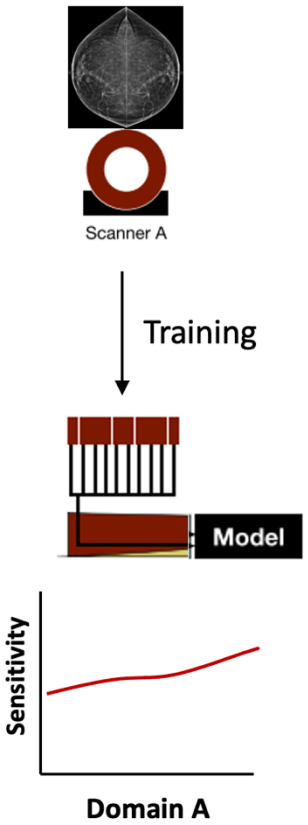
Clinical Problems

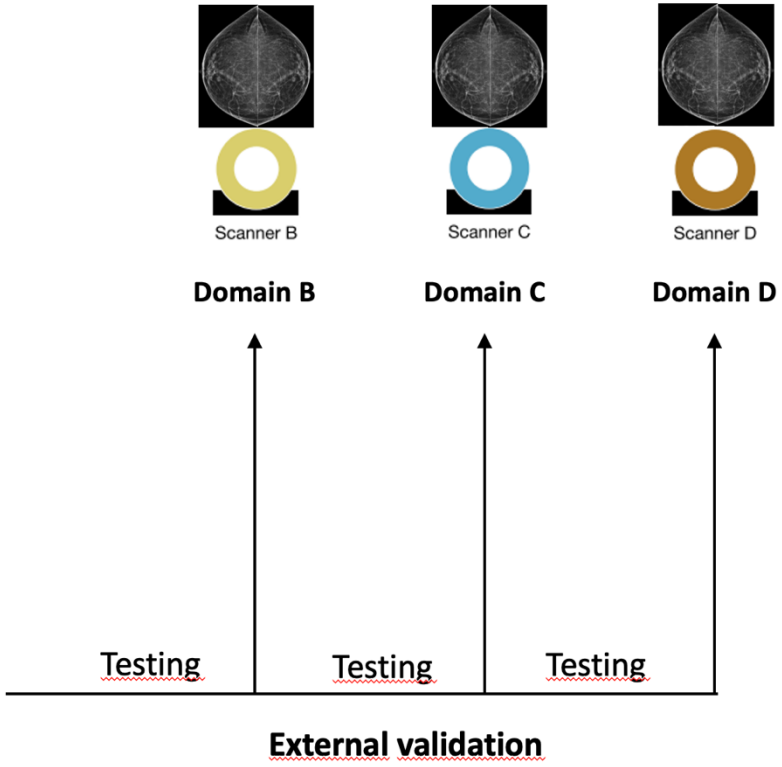
Clinical Problems

- **Problem**: Breast cancer detection in multi-center data
- **Input:** Mammography
- **Task**: Classification
- **Challenge**: Domain shift

Acquisition shift







Clinical Problems

- **Problem**: Diagnosis of Attention Deficit Hyperactivity Disorder (ADHD)
- **Input:** Anatomical and functional MRI
- **Task**: Analysis of brain structure/function
- **Challenge**: Signal registration

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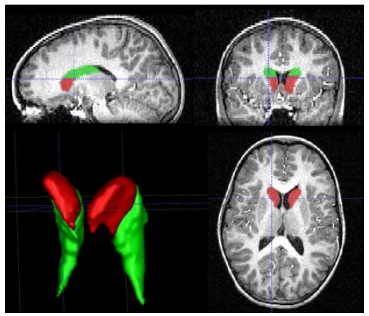
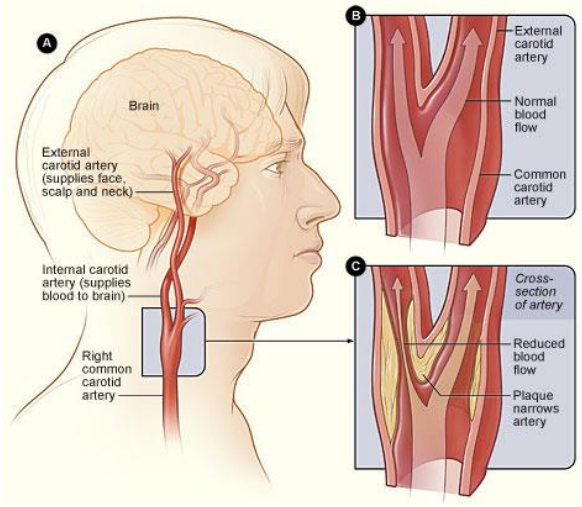


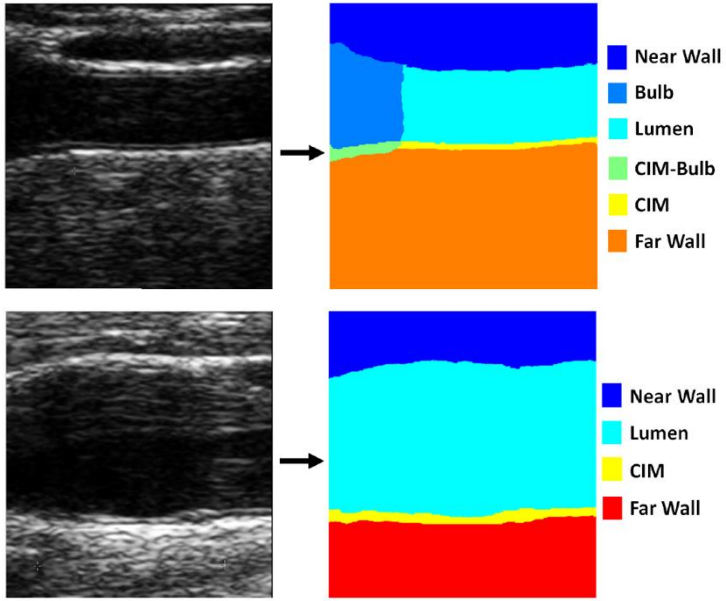
Departament de Psiquiatria i Medicina Legal
UAB

Neuroanatomical abnormalities in pediatric ADHD: Diminished volume of the **caudate nucleus.**

Clinical Problems

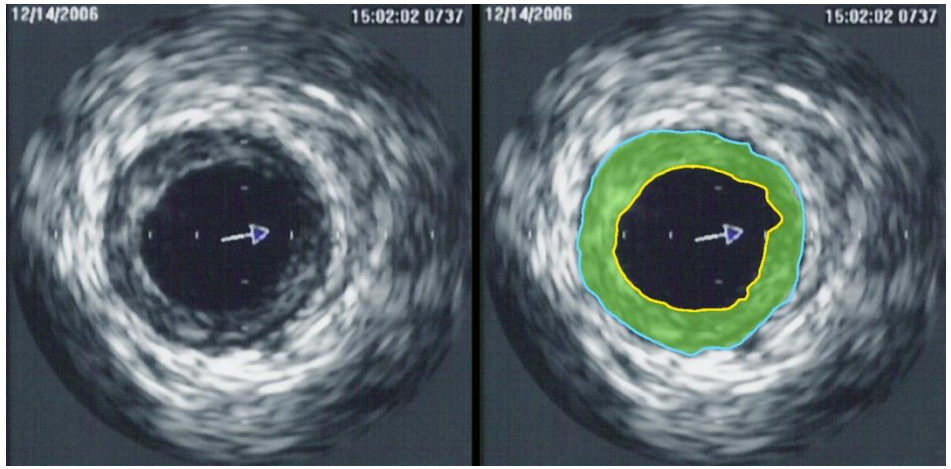
- **Problem**: Atherosclerotic plaque Detection and characterization. Cardiovascular events prevention.
- **Input:** Carotid artery ultrasound images
- **Task**: Segmentation and detection
- **Challenge**: Not enough data

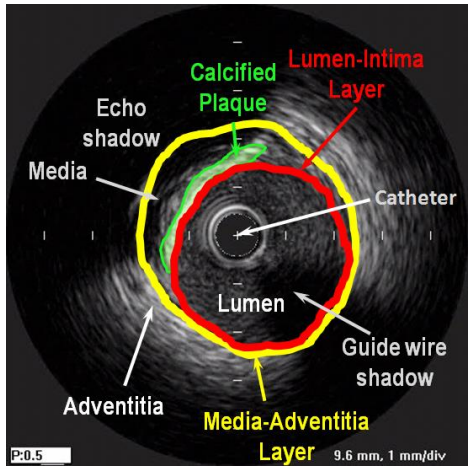




Clinical Problems

- **Problem**: Estimation of the stenosis percentage
- **Input: I**ntravascular ultrasound.
- **Task**: Segmentation

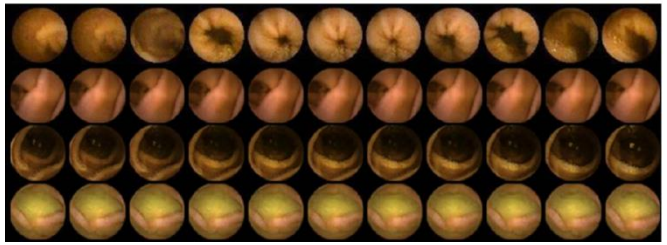




Clinical Problems

- **Problem**: Analysis of intestinal motility.
- **Input:** Wireless capsule endoscopic images
- **Task**: Contraction detection. Lession detection.
- **Challenge**: Not enough data

Wireless capsule endoscopy (WCE)



Frames from WCVE

From Problems to Methods

- We have problems

→ we need computational solutions.

Medical Image Processing and Analysis

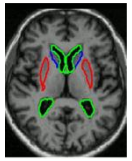
Medical Image Processing and Analysis

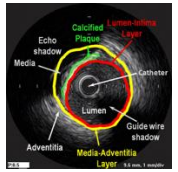
Image analysis is the extraction of meaningful information from images.

- Necessary to solve clinical problems.
- Interdisciplinary field.
- Many **automatic** techniques for different image analysis tasks.
- Focus on *understanding* & diagnosis of illness

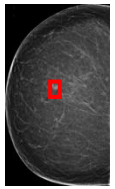
Medical Image Processing and Analysis

- **Pre-processing**: preparing raw signals for analysis.
  - Noise reduction (denoising),
  - Bias field correction in MRI
  - Intensity normalization.
- Medical image analysis **core tasks**:
  - Segmentation

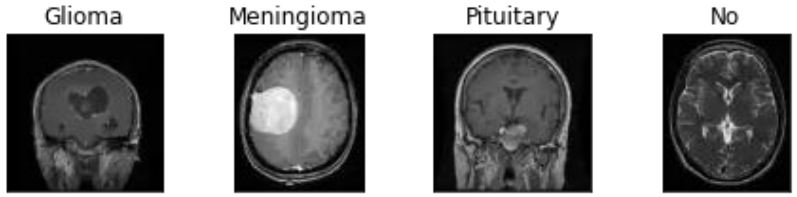




  - Detection



  - Classification

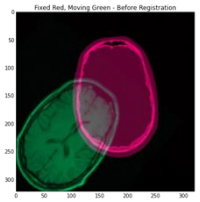


Manual procedure:
- Complex
- Time consuming
- Subjective

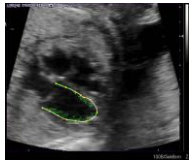
Automatic procedure
- Quick
- Replicable
- Feasible for large datasets

Medical Image Processing and Analysis

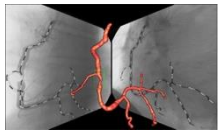
- Medical image analysis **additional tasks**:
  - Registration



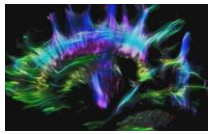
  - Tracking



  - Reconstruction

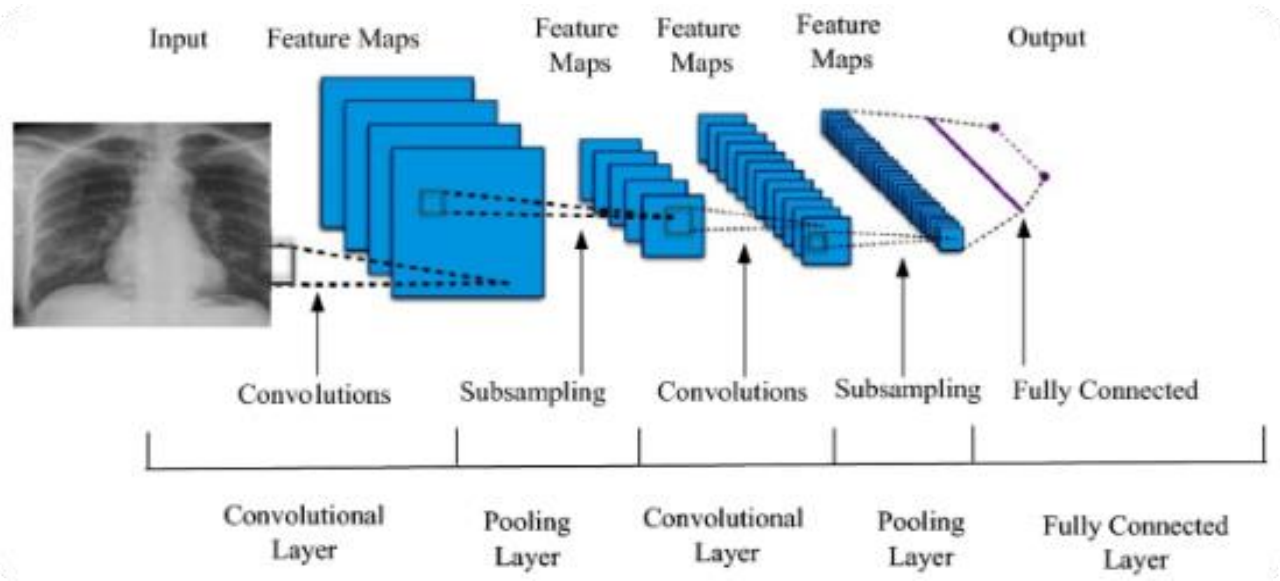


  - Visualization



The AI Revolution: Deep Learning

- Traditional computer vision relied on hand-crafted filters.
- Today, Convolutional Neural Networks (CNNs) learn features directly from data.

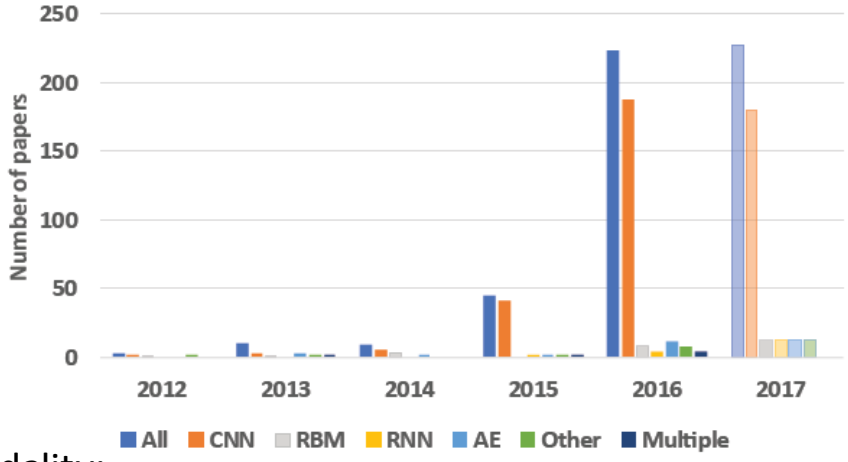


Source: www.mdpi.com

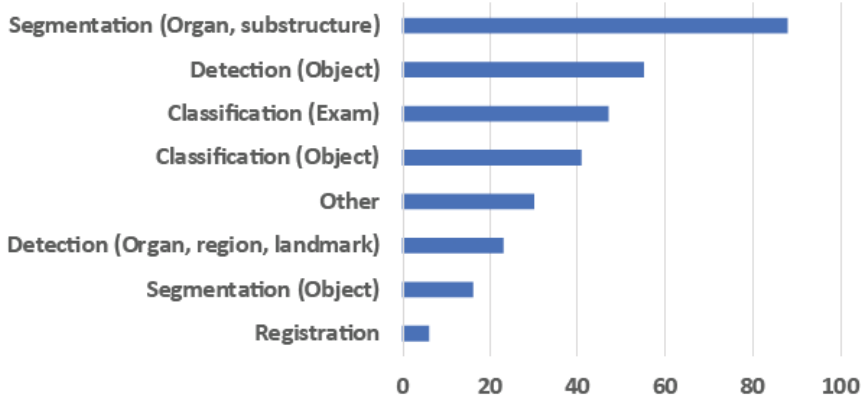
https://www.mdpi.com/information/information-16-00195/article_deploy/html/images/information-16-00195-g001.png

Deep Learning in Medical Image Analysis

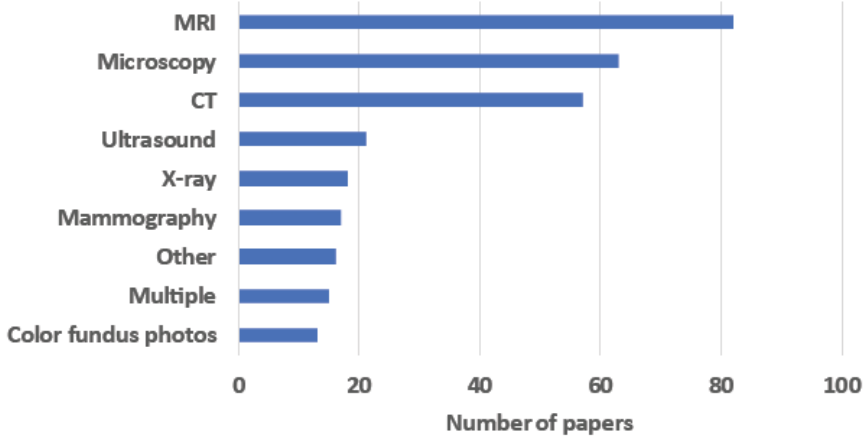
Total:



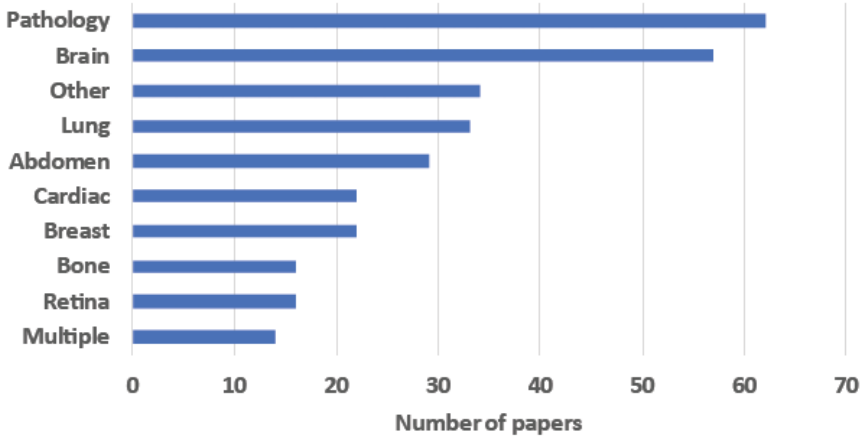
Task addressed:



Imaging modality:



Application area:



[Litjens et al., Arxiv January 2017]

Foundation Models

- One model → Many tasks.
- Example: BiomedCLIP
  - Uses text + images.
  - An AI "interpreter" that can analyze, classify, and search through medical images.
  - State-of-the-art performance in biomedical vision-language tasks.

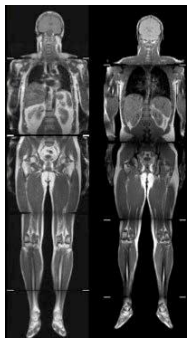
- Powerfull since:
  - Less labeled data needed
  - Generalization
  - Flexible

Example

- BioMedCLIP (Huggingface):
  https://huggingface.co/microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224/blob/main/biomed_clip_example.ipynb

Datasets

Medical imaging datasets



biobank

+

- Genetics
- Lifestyle
- Clinical records

- UK Biobank provides large-scale imaging data from 100.000 subjects (2014-2022).

- TCIA – **The Cancer Imaging Archive** consisting of extensive number of datasets as the Curated Breast Imaging Subset of Digital Database for Screening Mammography (CBIS-DDSM).

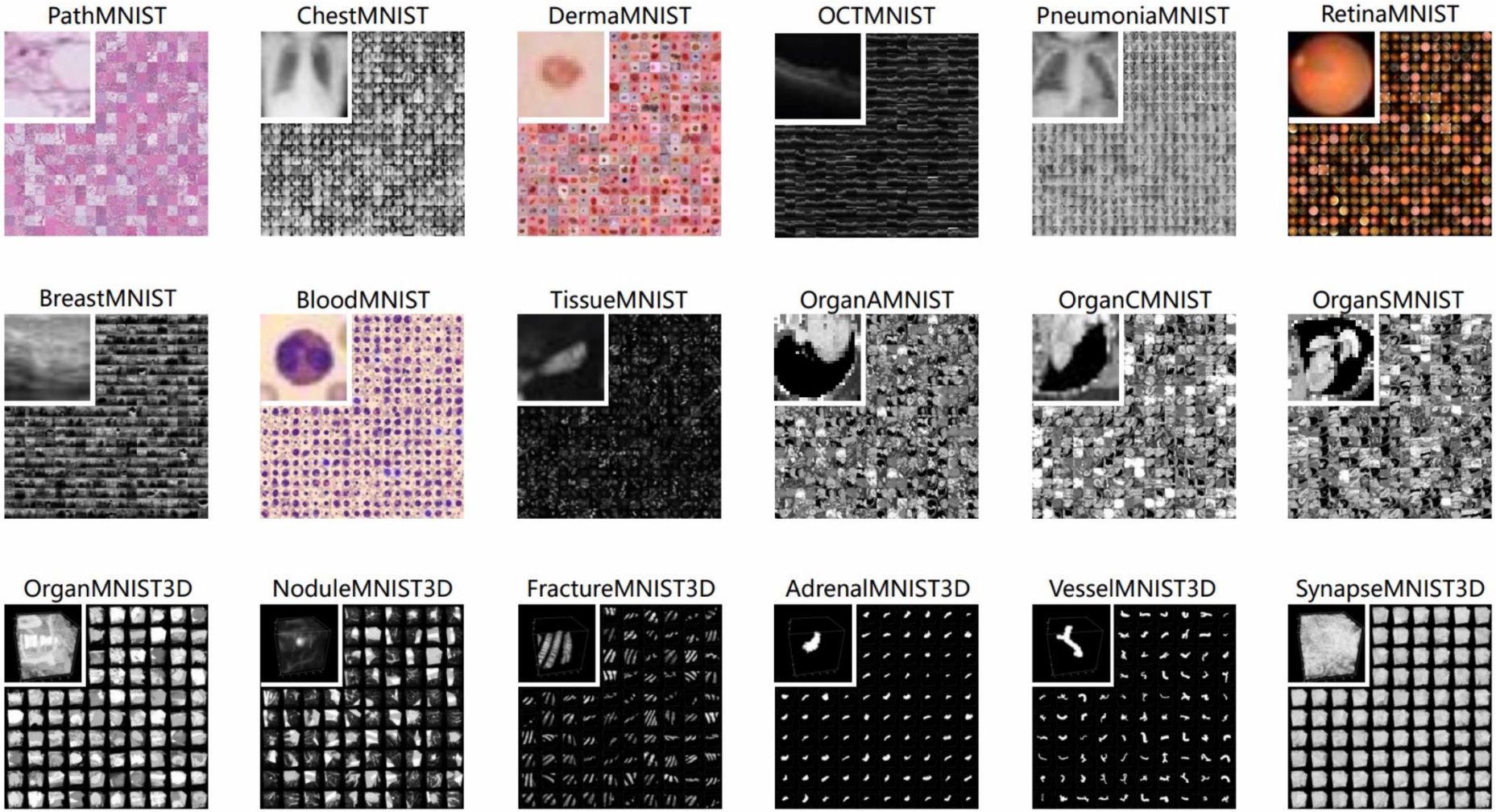
- Grand Challenge – data from over 100+ medical imaging competitions in data science

- Open Access Medical Imaging Datasets radRounds

Med MNIST

- Contains 12 datasets for 2D and 6 datasets for 3D. All images pre-processed into a small size of 28x28 (2D) or 28x28x28 (3D).
- Designed to perform classification

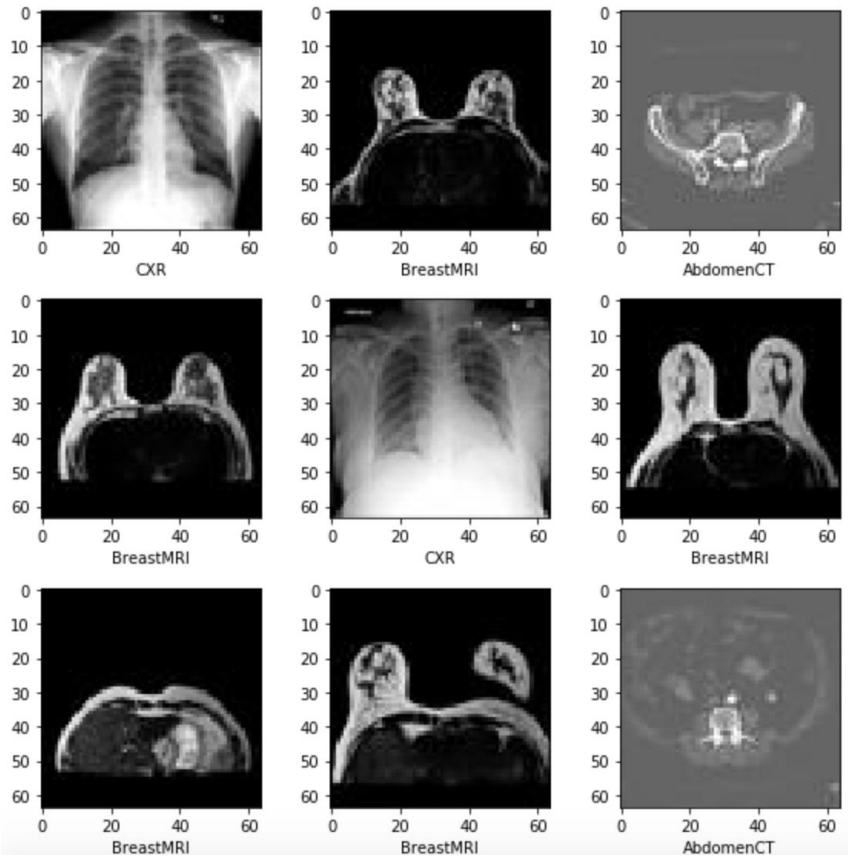
https://medmnist.com



Medical MNIST

- Medical MNIST: https://www.kaggle.com/datasets/andrewmvd/medical-mnist
- This dataset is a simple MNIST-style medical images in 64x64 dimension
- There are 58954 medical images belonging to 6 classes.
- Dessigned to multi-class classification

https://github.com/apolanco3225/Medical-MNIST-Classification/blob/master/MedNIST.ipynb



Tools

Medical Image Analysis Software

Open source software platforms for processing and analyzing medical images:

- 3D Slicer https://www.slicer.org/ (beginners).
- Insight Segmentation and Registration Toolkit (ITK) https://itk.org/ (coding).
- ITK-Snap http://www.itksnap.org/pmwiki/pmwiki.php (beginners).
- ImageJ (plug-in)

Take-home messages

- Medical imaging is essential for diagnosis and treatment.
- Medical images are diverse (and have different uses).
- Tasks define the problem
- Deep learning dominates today --> AI is transforming the field

References

References: DL Papers

- G. Litjens, T. Kooi, B. Ehteshami Bejnordi, et al. A Survey on Deep Learning in Medical Image Analysis (2017) arXiv:1702.05747v2.
- Lee, June-Goo et al. **Deep Learning in Medical Imaging: General Overview**. *Korean journal of radiology* vol. 18,4 (2017): 570-584. doi:10.3348/kjr.2017.18.4.570
- Tao Lei *et al.* Medical Image Segmentation Using Deep Learning: A Survey. (2020) arXiv:2009.13120v2
- Chan, HP., Samala, R.K., Hadjiiski, L.M., Zhou, C. (2020). **Deep Learning in Medical Image Analysis**. In: Lee, G., Fujita, H. (eds) Deep Learning in Medical Image Analysis . Advances in Experimental Medicine and Biology, vol. 1213. Springer. https://doi.org/10.1007/978-3-030-33128-3_1
- S. K. Zhou *et al*., A Review of Deep Learning in Medical Imaging: Imaging Traits, Technology Trends, Case Studies With Progress Highlights, and Future Promises, in *Proceedings of the IEEE*, vol. 109, no. 5, pp. 820-838, May 2021, doi: 10.1109/JPROC.2021.3054390. https://ieeexplore.ieee.org/document/9363915
- Varoquaux, G., Cheplygina, V. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digit. Med.* **5,** 48 (2022). https://doi.org/10.1038/s41746-022-00592-y
- Liu, X. et al. A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis. The Lancet Digital Health (2019). https://www.thelancet.com/journals/landig/article/PIIS2589-7500(19)30123-2/fulltext

More References

Interesting Links:

- Methods in (Bio)Medical Image Analysis (CMU Course): http://www.cs.cmu.edu/~galeotti/methods_course/
- Deep Learning & Medical Diagnosis: https://medium.com/data-science/deep-learning-medical-diagnosis-c04d35fc2830
- Guide to Semantic Segmentation with Deep Learning http://blog.qure.ai/notes/semantic-segmentation-deep-learning-review

Books:

- *Machine Vision*, Wesley E. Snyder & Hairong Qi
- *Insight into Images: Principles and Practice for Segmentation, Registration and Image Analysis*, Terry S. Yoo (Editor)

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